4. Common CNN Architectures

Task 4.1: Tiny-ImageNet with VGG16

Report:

- Fine-tuned VGG16 on Tiny-ImageNet. Best test accuracy reported in Table 1. Training/validation accuracy curves show convergence.

Table 1: Tiny-ImageNet: best test accuracy.

Metric	Value
Test Accuracy (%)	38.27

Task 4.2: ConvNeXt Model Scaling

Report:

- Compared ConvNeXt variants (tiny, small, base) on Tiny-ImageNet. Accuracy and MSE vs. number of parameters (Figure 1).

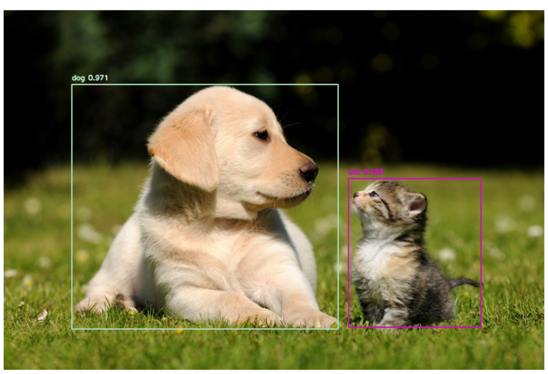

Figure 1: ConvNeXt scaling: accuracy and MSE vs. number of parameters.

5. RNN

Task 5.1: RNN Regression

Report:

- Window size sweep: RMSE vs. window size (Table 2). Best window size = 1 for lowest test RMSE. Test curve plot in Figure 2.

Table 2: RMSE vs. window size for LSTM regression.

Window Size	Train RMSE	Test RMSE
1	23.22	50.82
3	22.28	56.25
6	21.90	57.57
12	17.68	81.94

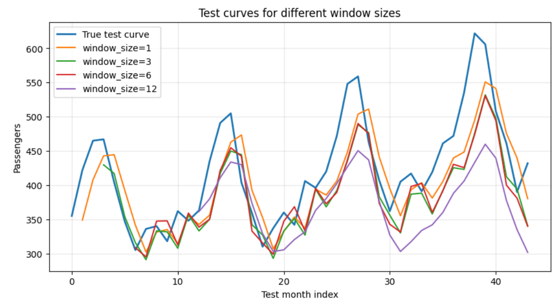

Figure 2: Test curves for different window sizes (Task 5.1).

Task 5.2: Text Embeddings

Report:

- Compared Embeddings, LSTM, and LSTM+GloVe models. Test accuracy and sentiment scores (Table 3). Accuracy/loss curves in Appendix.

Table 3: Text embedding models: test accuracy and sentiment scores.

Model	Test Acc (%)	Neg Score	Pos Score
Embeddings	78.00	0.48	0.48
LSTM	73.28	0.25	0.25
LSTM+GloVe	78.21	0.34	0.66

Task 5.3: Text Generation

Report:

- BLEU score vs. temperature for char-level and word-level models (Table 4, Figure 3).

Table 4: BLEU score vs. temperature.

Temperature	Char BLEU	Word BLEU
0.0	0.0144	0.0116
0.5	0.0177	0.0111
1.0	0.0133	0.0115
2.0	0.0115	0.0078

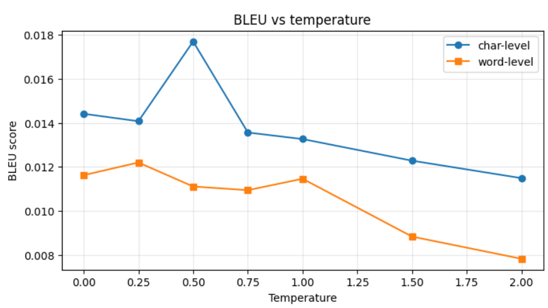

Figure 3: BLEU vs. temperature for char-level and word-level generation.

6. Autoencoders

Task 6.1: Representation Learning

Report:

- Autoencoder for representation learning. Classifier accuracy and MSE for linear AE, conv AE, and PCA (Table 5).

Table 5: Representation learning: classifier accuracy and MSE.

Model	Val Acc (%)	MSE
Linear AE	81.30	0.026
Conv AE	81.69	0.029
PCA (10 comp)	81.38	0.0256

Task 6.2: Custom Loss Functions

Report:

- Effect of custom loss terms on denoising MSE (Table 6, Figure 4).

Table 6: Denoising MSE for different loss functions.

Loss	Test MSE
MSE	–
MAE	–
SSIM	–

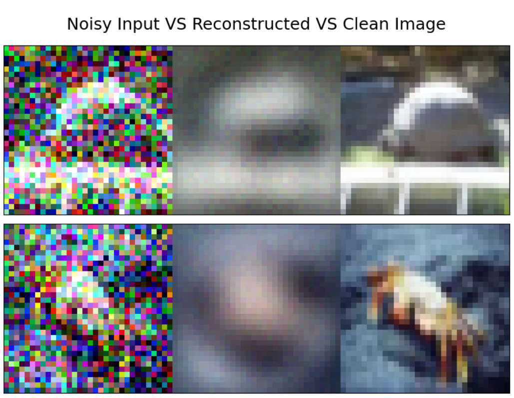

Figure 4: Denoised image examples for different loss functions.

7. VAE GAN

Task 7.1: MNIST Generation (VAE/GAN)

Report:

- VAE: reconstruction MSE and Inception Score. GAN: Inception Score. Table 7.

Table 7: VAE vs. GAN: MSE and Inception Score on MNIST.

Model	MSE	Inception Score
VAE (latent=10 + KL)	10.983575	5.342967
VAE (no KL)	8.836834	2.374818
GAN (latent=10)	NaN	1.000000

Task 7.2: MAE vs. cGAN Colourisation

Report:

- Mean MAE for MAE and cGAN on CIFAR-10 colourisation (Table 8). Qualitative examples in Appendix.

Table 8: MAE vs. cGAN: mean MAE on CIFAR-10 colourisation.

Model	Mean MAE
MAE	0.0899
cGAN	0.0970

8. RL

Task 8.1: Q-learning

Report:

- Q-learning (and SARSA) with ϵ-greedy and Softmax policies. Average reward (last 50 episodes) vs. training episodes (Figure 5).

9. Diffusion Model

Task 9.1: Forward Noising

Report:

- Forward noising: clean image, noised image, and noise (Figure 6). Training loss curve in Appendix.

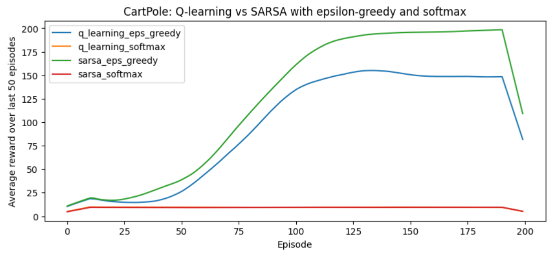

Figure 5: Q-learning: average reward (last 50 episodes) vs. training episodes.

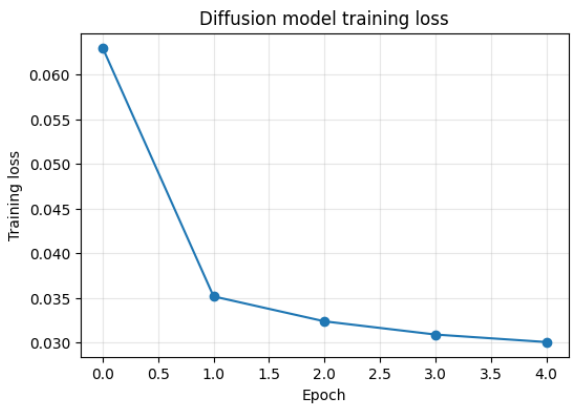

Figure 6: Diffusion forward process: clean image, noised image, and noise.

10. Data Classification YOLO

Task 10.1: Roof Classification

Report:

- Finetuned `yolov8n-cls.pt` on aerial roof images (flat vs. pitched). Dataset: course-provided aerial data merged with Zenodo BRAILS roof-type images.
- Data augmentation: RandAugment, CutMix, MixUp, Random Erasing.
- Training: epochs=30, batch=64, imgsz=320, patience=10. Best checkpoint at epoch 26.
- Test accuracy and learning curves (Figure 7, Table 9).

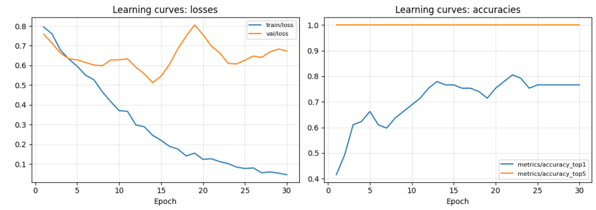

Figure 7: YOLOv8 roof classification: training loss and accuracy curves.

Table 9: YOLOv8 roof classification: test metrics.

Metric	Value
Test Top-1 Accuracy (%)	83.61
Test Top-5 Accuracy (%)	100.0
Best checkpoint	epoch 26

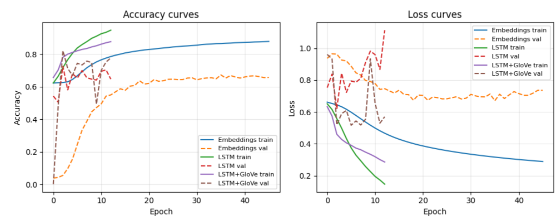

Figure 8: Accuracy and loss curves for Embeddings, LSTM, and LSTM+GloVe models.

Appendix

Task 5.2: Accuracy and loss curves

Task 7.2: MAE vs. cGAN qualitative examples

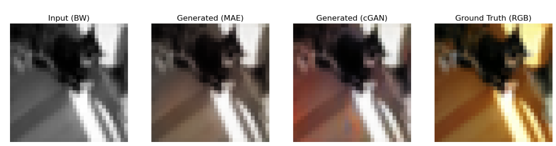

Figure 9: MAE vs. cGAN colourisation on CIFAR-10.

Task 9.2: Diffusion training loss

Figure 10: Diffusion model training loss vs. epochs.